

3. Row 2 Column 1 = Row 3 Column 1 = 0

Row 3 Column 2 = 0

∴ The condition that each pivot column contains only 0 below the pivot position is True. $\square$

∴ The matrix $\begin{bmatrix} 1 & 0 & 2 & -5 \\ 0 & 5 & 4 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ is in echelon form.

There exists a row in the matrix of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. By Theorem 2, the system is inconsistent, i.e. no solution set exists.

∴ b is not a linear combination of a_1, a_2 and a_3 . $\square$

In Exercises 13 and 14, determine if b is a linear combination of the vectors formed from the columns of the matrix A .

13. $A = \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 5 \\ -2 & 8 & -4 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ -7 \\ -3 \end{bmatrix}$

Solution: We have to determine if $x_1 \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} + x_2 \begin{bmatrix} -4 \\ 3 \\ 8 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 5 \\ -4 \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \\ -3 \end{bmatrix}$, for some $x_1, x_2, x_3 \in \mathbb{R}$.

This is equivalent to finding out whether the augmented matrix $\begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{bmatrix}$ has a solution set.

$$\begin{aligned} &\begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{bmatrix} \xrightarrow[\text{Interchange Row 2 and Row 3}]{\text{Interchange Row 2 and Row 3}} \begin{bmatrix} 1 & -4 & 2 & 3 \\ -2 & 8 & -4 & -3 \\ 0 & 3 & 5 & -7 \end{bmatrix} \xrightarrow[\text{+2} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 0 & 0 & 3 \\ 0 & 3 & 5 & -7 \end{bmatrix} \end{aligned}$$